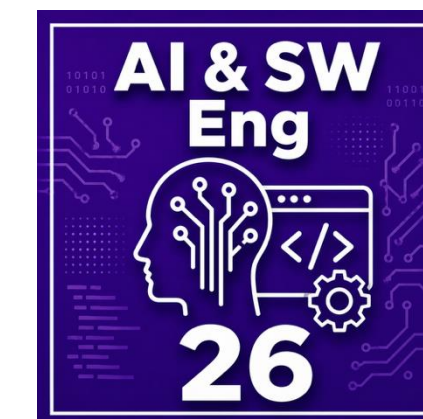




Higher Diploma in Science in Computer Science
Certificate in Computer Science
Certificate in Software Engineering
Certificate in Artificial Intelligence & Machine Learning

Overview of Structure & Purpose of the **Online** Materials

Tutors



Slack



Moodle



Youtube



4 Key Web Services

Tutors



Github ID

Slack



personal email
> setu email

Moodle



setu email

Youtube



open
> setu email

Programme homepage

<https://tutors.dev/course/setu-cert-comp-sci-2026>



  **Certificate in Computer Science 2026**
Department of Computing & Mathematics, SETU

 Current Week
0

 **Layout**



 >  Certificate in...

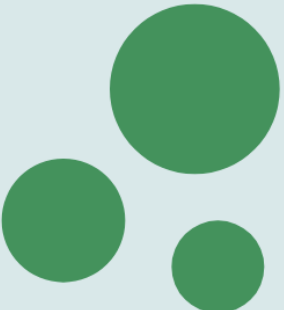
 

Semester One - September 2026 - January 2027



FAQ 

Questions and Answers
for new students on the
programme

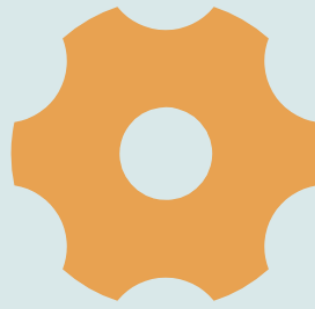
Induction 

induction · structure ·
schedules · handbook

Life etc... 

time · space · balance ·
experiences · tips ·
techniques

**Web
Development I** 

html · css · layout · web
apps · web frameworks ·
templating

Programming 

algorithms · data
structures · processing ·
python · libraries

**Semester 1
Schedule** 

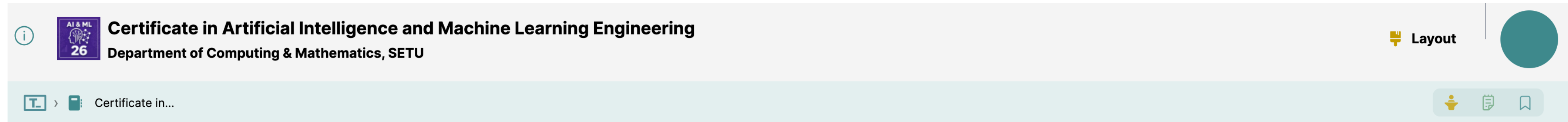
timetables · semester
calendars · assessment
schedules

Links to individual module home pages

Programme homepage



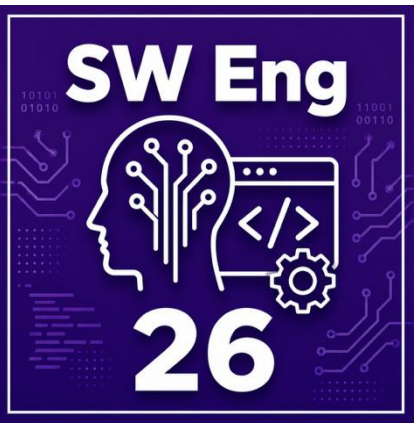
<https://tutors.dev/course/setu-cert-ai-ml-2026>



Links to individual module home pages

Programme homepage

<https://tutors.dev/course/setu-cert-sw-eng-2026>



**Certificate in Software Engineering**
Department of Computing & Mathematics, SETU

Layout



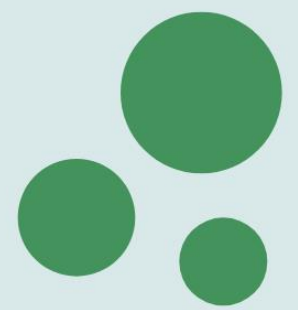
 >  Certificate in ...

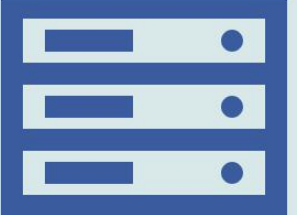


Semester One

Academic Calendar
weeks · reading week · semester · onsites

FAQ
Questions and Answers for new students

Induction
induction · structure · schedules · handbook

Web App Development
javascript · node · apis · tdd · frameworks · front-end · sveltekit · ...

Developer Operations
cloud computing · scripting · scaling · automation · monitoring

Life etc...
time · space · balance · experiences · tips · techniques

Links to individual module home pages

A course/module homepage

1 topic/card for each week

Programming Fundamentals

Siobhan Roche, Mairead Meagher, Pete Windle

Current Week

Semester 1

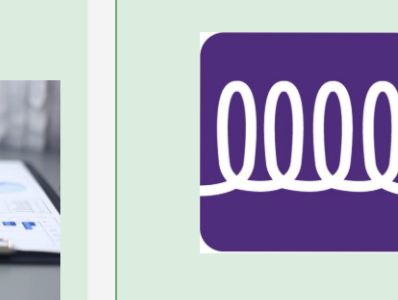
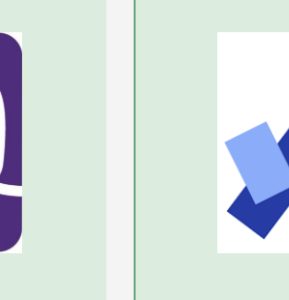
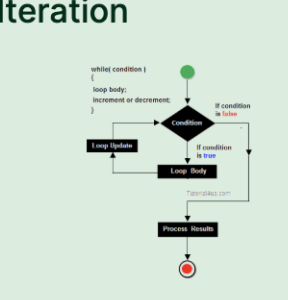
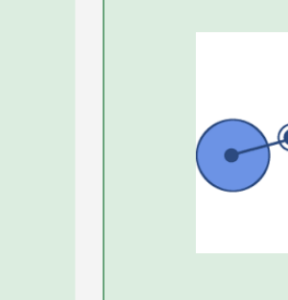
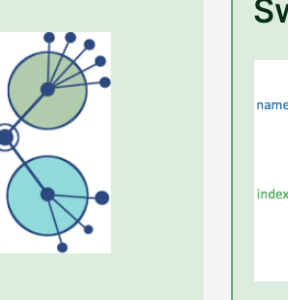
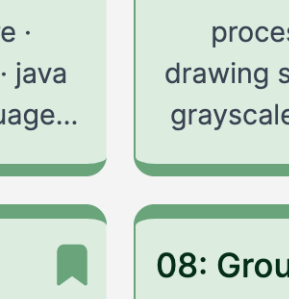
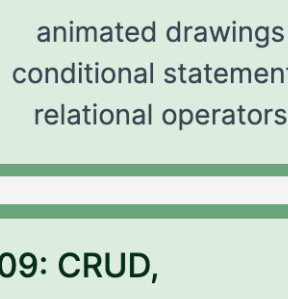
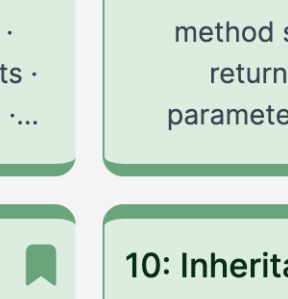
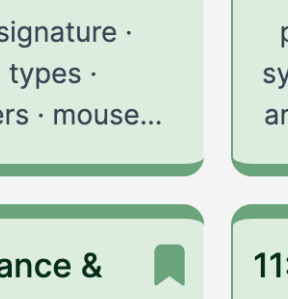
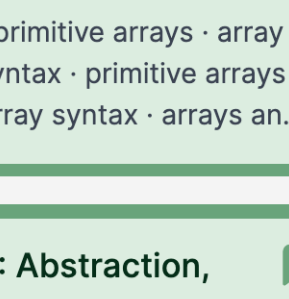
Search

Layout

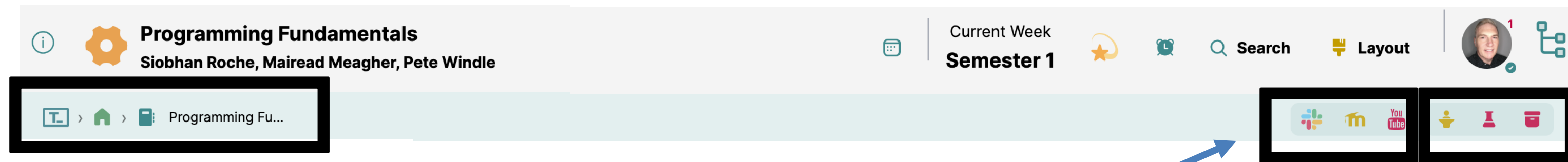
>

>

Programming Fu...

00: Induction	01: First Contact	02: Selection & Iteration	03: Methods	04: Arrays and Swing	05: IntelliJ and Basic I/O
 Assignment Specifications This section details the assignments for module	 module structure · delivery approach · java programming language...	 processing · java · drawing shapes · colour · grayscale · RGB · synta...	 animated drawings · conditional statements · relational operators ·...	 method signature · return types · parameters · mouse...	 primitive arrays · array syntax · primitive arrays · array syntax · arrays an...
06: Classes	07: Shop Validation and...	08: Grouping Objects	09: CRUD, Persistence,...	10: Inheritance & Misc	11: Abstraction, Interfaces,...
 String methods · Objects · Classes · behaviour · attributes · Bank...	 IntelliJ · Scanner · OO recap · ShopV1	 primitive arrays · classes · algorithms for collections · ArrayLists	 CRUD - persistence · exceptions · debugging · misc	 Utilities · Debugger · Inheritance	 Abstract classes, methods, interfaces, sorting, searching
12: Collections and Testing					
 HashMap, HashSet and JUnit					

Navigation Bar - 1



Breadcrumb Navigation

Companions

Wall

Links to:

- course slack channel,
- moodle module,
- YouTube playlist,
- other

- Talks
- Labs
- Archives

Navigation Bar - 2

The screenshot shows the Blackboard course navigation bar for the course "Programming Fundamentals". The bar includes the course title, instructors (Siobhan Roche, Mairead Meagher, Pete Windle), the current week (Semester 1), and navigation links for Search, Layout, and a user profile. Blue arrows point from the navigation bar to labels below: "Course Info" points to the information icon, "Calendar" points to the calendar icon, and "Layout Options" points to the layout icon.

Course Info

Calendar

Layout Options

Semester 1	
Type	Date Starts
-3	Dec 23
-2	Dec 30
-1	Jan 6
0	Jan 13
1	Jan 20
2	Jan 27
3	Feb 3
4	Feb 10
Reading 1	Feb 17
5	Feb 24
6	Mar 3
7	Mar 10
Reading 2	Mar 17
8	Mar 24
9	Mar 31
10	Apr 7
Easter 1	Apr 14
Easter 2	Apr 21
11	Apr 28
12	May 5
Reading 5	May 12
Reading 6	May 19

Mood

-  neutral 
-  fine
-  delighted
-  confident
-  overwhelmed
-  confused
-  drained

Tutors Live

Full Text Search

Layout Options

Appearance



Card Style



Theme



Code Style



Layout



Language



Guided Tour

 [Start Tour](#)

Table of Contents

▼  01: First Contact

- ▼  1a: Introduction
 -  Online Class 20251
 - ▼ 
 - 
 - 
 - 
 - 
 -  Quiz
 -  INTRODUC
 -  Introduction
 -  Static Draw
 -  Formatting
 -  Lab-1a
- ▼  1b: Animating
 -  Online Class (Wed 22-07)
 - 
 - ▼ 
 - 
 - 
 -  Quiz
 -  INTRODUC
 -  Basic Anima
 -  Data Types
 -  Lab-1b

▼  02: Selection & It

- ▼  2a: Selection

Theme Support



TOPICS

1 topic/card for each week



Programming Fundamentals

Siobhan Roche, Mairead Meagher, Pete Windle

 Current Week
Semester 1





 Search

 Layout

¹

 >  >  Programming Fu...





Topic	Sub-topics
00: Induction	module structure · delivery approach · java programming language...
01: First Contact	processing · java · drawing shapes · colour · grayscale · RGB · synta...
02: Selection & Iteration	animated drawings · conditional statements · relational operators · ...
03: Methods	method signature · return types · parameters · mouse...
04: Arrays and Swing	primitive arrays · array syntax · primitive arrays · array syntax · arrays an...
05: IntelliJ and Basic I/O	IntelliJ · Java Virtual Machine (JVM) · main method · Scanner
06: Classes	String methods · Objects · Classes · behaviour · attributes · Bank...
07: Shop Validation and...	IntelliJ · Scanner · OO recap · ShopV1
08: Grouping Objects	primitive arrays · classes · algorithms for collections · ArrayLists
09: CRUD, Persistence,...	CRUD - persistence · exceptions · debugging · misc
10: Inheritance & Misc	Utilities · Debugger · Inheritance
11: Abstraction, Interfaces,...	Abstract classes, methods, interfaces, sorting, searching
12: Collections and Testing	HashMap, HashSet and JUnit

Single Topic

e.g. Week 1

01: First Contact
Programming Fundamentals

Current Week
Semester 1

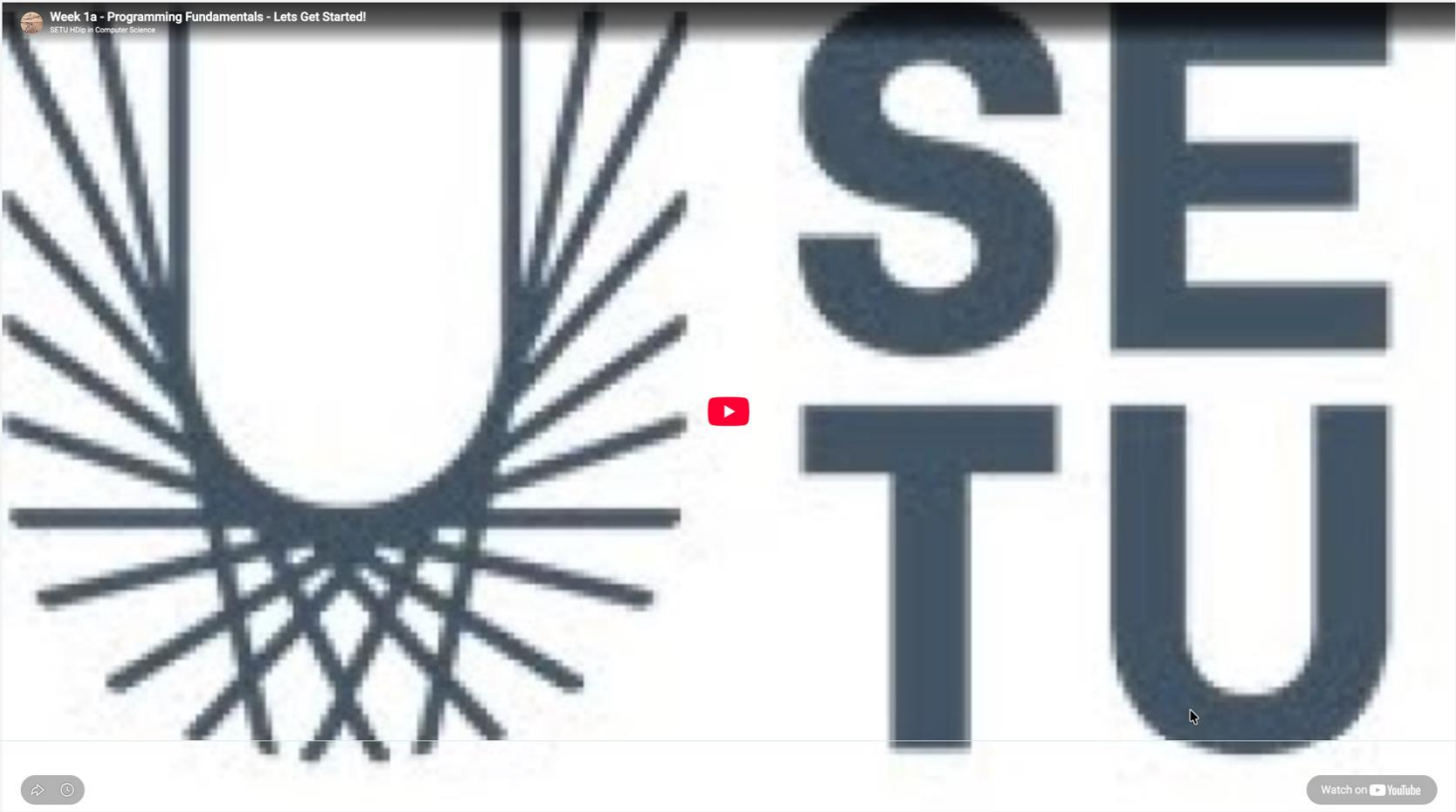
Search

Layout

Programming Fu... | 01: First Conta...

1a: Introduction to Processing

Week 1a - Programming Fundamentals - Lets Get Started!



Watch on YouTube

Online Class Monday 20-01-2025
12:15 PM

INTRODUCTION
Intro to week 1

Introduction to the PDE
Processing - Processing Development Environment (PDE).

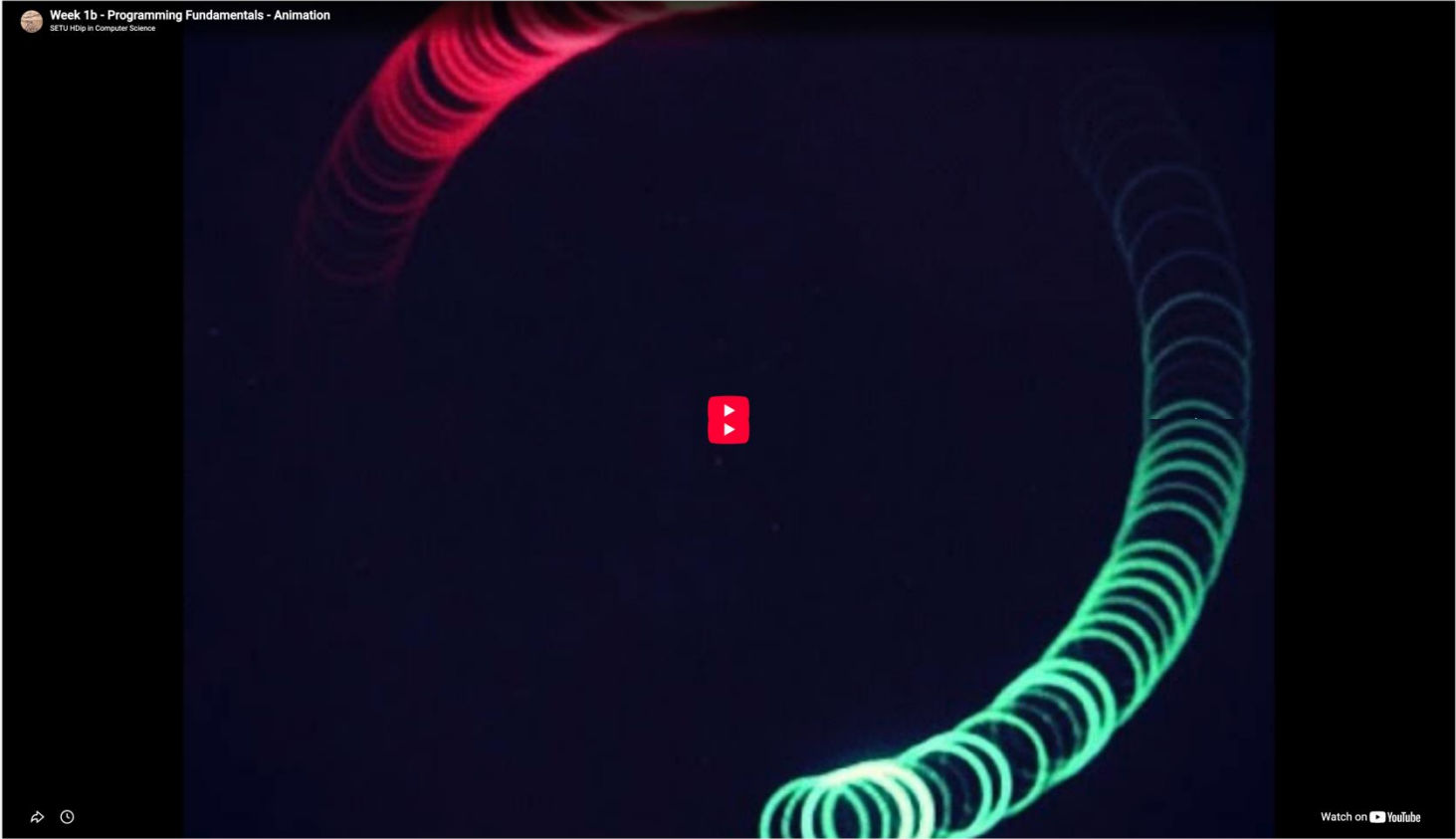
Static Drawings
coding - static drawings - basic shapes - colour - grayscale - RGB

Formatting Shapes
fill - stroke - commenting code.

Lab-1a
static drawings - basic shapes - colour - grayscale - RGB

1b: Animating your Drawings

Week 1b - Programming Fundamentals - Animation



Watch on YouTube

Online Class Week 1 Talk b (Wed 22-01-2025 12:15 PM)
Online Class Week 1 Talk b (Wed 22-01-2025 12:15 PM)

INTRODUCTION
recap week 1 talk 1 * introduce talk 2

Basic Animation
setup() - draw() - animate drawings - system variables

Data Types
primitive data types - variables - arithmetic operators

Lab-1b
animate simple drawings - variables - system variables - primitive data types - ...

Topic Video

Pre-recorded or live stream
(e.g. Monday’s Class)

Sub Topic 1

Talks/slides/pdf & Labs

Topic Video

Pre-recorded or live stream
(e.g. Wednesday’s Class)

Sub Topic 2

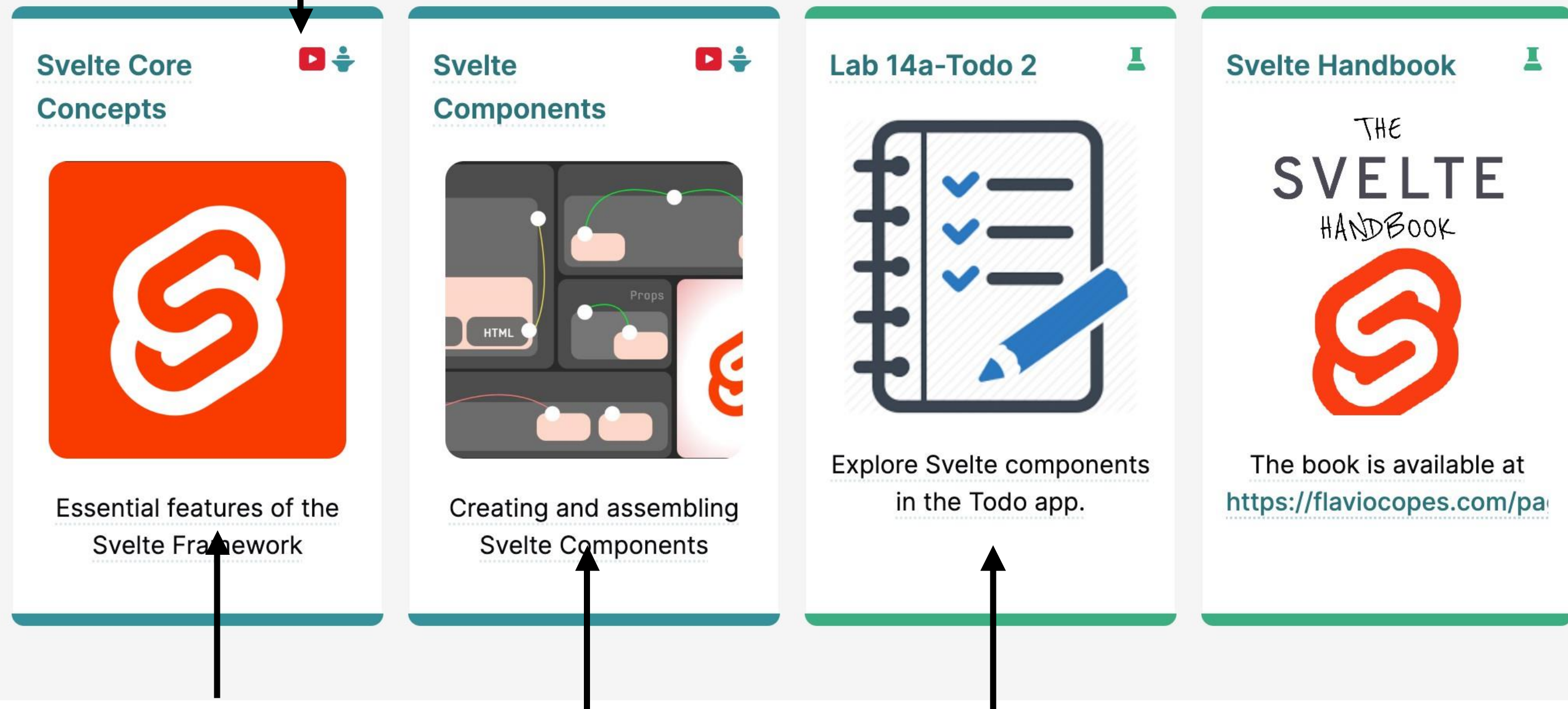
Talks/slides/pdf & Labs

Sub Topic

Svelte Fundamentals



Single
Video



Talk Slide PDFs

Lab

Talk

Donate in Svelte
Full Stack Web Development

Search

Layout

Full Stack Web ...

16: Svelte Appl...

Svelte Applicat...

Donate in Svelt...

4 of 18

Context

The context API is provided by 2 functions which are provided by the svelte package: `getContext` and `setContext`

You set an object in the context, associating it to a key

In another component you can use `getContext` to retrieve the object assigned to a key:

➔ **Context:** A feature to allow a component communicate with multiple descendants

```
<script>
import { setContext } from 'svelte'

const someObject = {}

setContext('someKey', someObject)
</script>
```

```
<script>
import { getContext } from 'svelte'

const someObject = getContext('someKey')
</script>
```

4

16: Svelte Applications



- Svelte Applications
- Svelte Applications
- Why Svelte?
- Donate API
- Donate in Svelte
- Lab-16a-Donate-Hapi-3
- Lab-16b-Donate-Svelte-3

Embedded
PDF reader
custom
with
Navigation

Talk
Context-
related
talks &
videos

Labs

Lab Steps



- Objectives
- Create
- Router**
- Components
- Pages
- Donate & Report
- Login & Signup
- Solutions

Router

Like web apps, front end applications also have the concept of a router. Svelte itself does not provide one, but there are several popular options:

- <https://www.routify.dev/>
- <https://github.com/AlexxNB/tinro>
- <https://github.com/ItalyPaleAle/svelte-spa-router>

We will use the last one - which is documented in detail in this svelte text book:

- [Svelte 3 Up & Running](#)

Start by installing it:

```
npm install svelte-spa-router
```

To try it out, create three svelte file in a new **pages** folder in the project:

pages/Main.svelte

```
<div class="notification is-link">
  Hello from Main!
</div>
```

pages/Login.svelte

```
<div class="notification is-info">
  Hello from Login!
</div>
```

pages/Signup.svelte

```
<div class="notification is-warning">
  Hello from signup!
</div>
```

This is a revised App.svelte, to include the routing table:

App.svelte

Markdown
formatted text

Code block
syntax
highlighting

Search

Enter search term:

env

<pre>"env": {</pre> Lab-01-Playtime-0-1-0/ Code Quality Tools	<pre># dotenv environment variables file</pre> Lab-01-Playtime-0-1-0/ Code Quality Tools	<pre># dotenv environment variables file</pre> Lab-01-Playtime-0-1-0/ Code Quality Tools
<pre>.env</pre> Lab-01-Playtime-0-1-0/ Code Quality Tools	<pre>.env.test</pre> Lab-01-Playtime-0-1-0/ Code Quality Tools	<pre>.env.production</pre> Lab-01-Playtime-0-1-0/ Code Quality Tools
In general we should attempt to "externalise" sensitive information, secrets or API details to a text file. This is usually called <code>.env</code> Lab-02-Playtime-0-2-0/ Exercises	.env Lab-02-Playtime-0-2-0/ Exercises	https://github.com/motdotla/dotenv#readme Lab-02-Playtime-0-2-0/ Exercises
In general we should attempt to "externalise" sensitive information, secrets or API details to a text file. This is usually called <code>.env</code> Lab-03-Playtime-0-3-0/ Exercise 3	.env Lab-03-Playtime-0-3-0/ Exercise 3	https://github.com/motdotla/dotenv#readme Lab-03-Playtime-0-3-0/ Exercise 3
<pre>npm install dotenv</pre> Lab-03-Playtime-0-3-0/ Exercise 3	Import the package in server, and check for the presence of the .env file, exiting if not found: Lab-03-Playtime-0-3-0/ Exercise 3	<pre>import dotenv from "dotenv";</pre> Lab-03-Playtime-0-3-0/ Exercise 3
<pre>import dotenv from "dotenv";</pre> Lab-03-Playtime-0-3-0/ Exercise 3	<pre>const result = dotenv.config();</pre> Lab-03-Playtime-0-3-0/ Exercise 3	If we have included the <code>.env</code> file above at the root of our project, then we can directly access its entries. So, our cookie configuration can now import these values:

Youtube Live

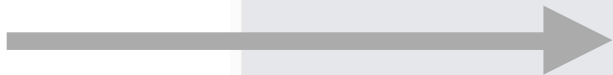


Live Webinars
Delivery



Live Stream

Live stream of
webinar for given
topic



2021 Computer Systems and Networks

Device/Service Discovery



- Automatic detection of devices and services on a computer network(usually a LAN).
- Requires a common language for interoperability
- Many different protocols, based on physical layer implementations and domains:
 - Dynamic Host Configuration Protocol (DHCP)
 - Bluetooth Service Discovery Protocol (SDP)
 - Universal Description Discovery and Integration (UDDI)
 - [Simple Service Discovery Protocol \(SSDP\)](#)

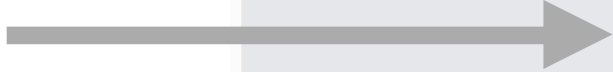
MORE VIDEOS

4:33 / 1:36:38

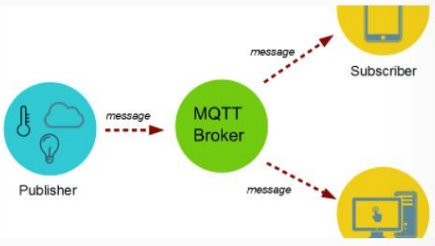
YouTube

07: Application Layer: Messaging

Slides & Labs for
the topic

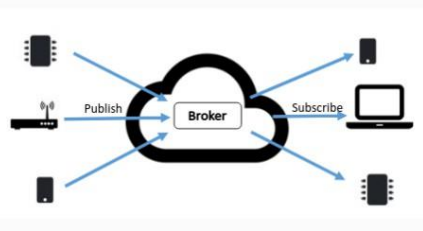


MQTT and HTTP



Data Representation · Publish Subscribe · Web APIs

Lab Messaging



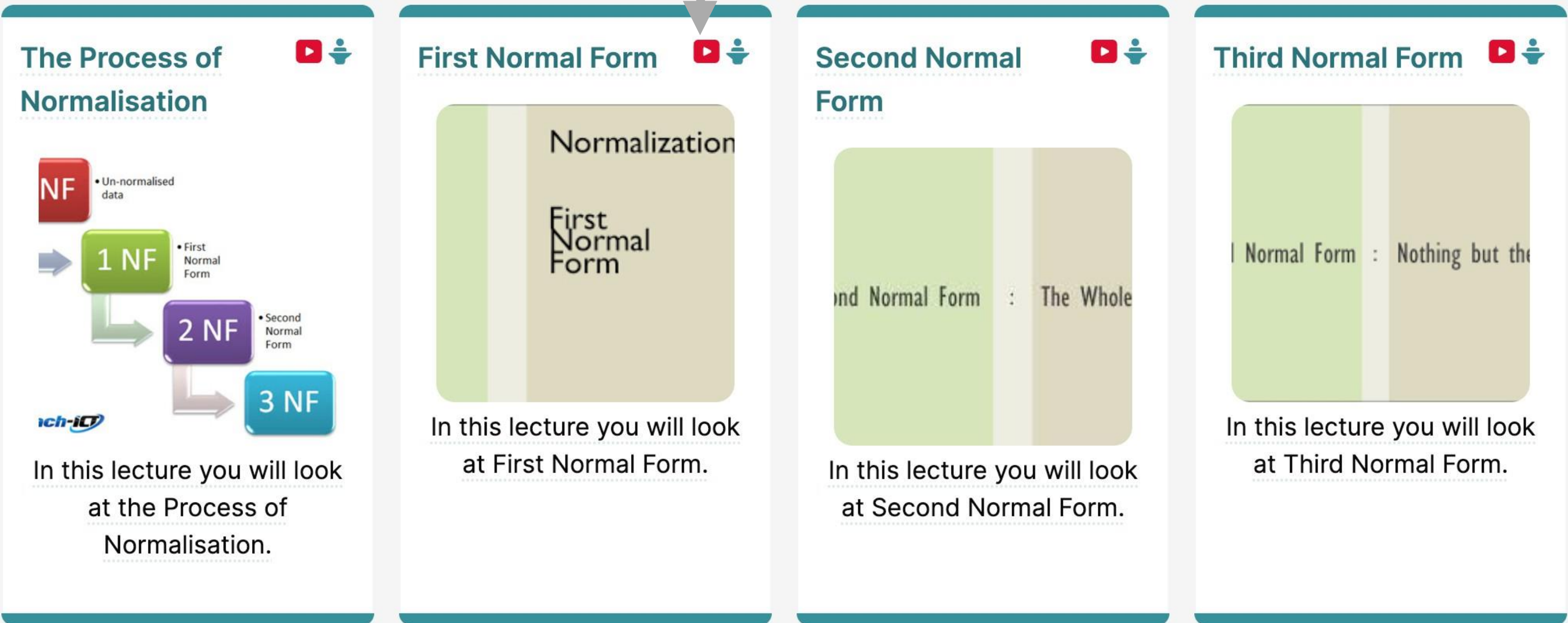
Use MQTT to connect a device



Pre-recorded Videos for selected slides & labs / may also be clips from the live talk after it finishes



Stages of Normalisation



Moodle



SETU E-Learning Platform

Used for Assignments
(Submission + feedback + grades + choice)





NewsMy HomeCalendar

3

My Home

+ Add a block

Recently accessed modules



★ Higher Diploma in Computer S...
Non-Banner Modules



★ Sample Course Layout: One
Moodle Training

Profile

Grades

Calendar

Messages

Private files

Reports

Preferences

Log out

Grades

Higher Diploma in Computer Science

[My Home](#) / [My modules](#) / [Higher Diploma in Computer Science](#)

Programming



-  [Assignment 1](#)
-  [Assignment 2](#)
-  [Assignment 3](#)

Web Development



-  [Assignment 1](#)
-  [Assignment 2](#)

Skills Studio 1



-  [Assignment](#)

Assignment
Submissions



Assessment feedback

Web Development



 Assignment 1

 Assignment 2

Assignment: Submit

https://moodle.wit.ie/mod/assign/view.php?id=2407050&rownum=0&action=grader&userid=42131

Module: Higher Diploma in Computer Science

Assignment: Submit

View all submissions

 @mail.wit.ie

Change user

Due date: 11 March 2018, 11:55 PM

33 of 39

Well done  his for a excellent effort - your grade was 74%

The structure of the project was excellent, with effective use of file and folder names. Indentation was good (although head and body were not indented) and the use fo semantic elements was spot on. You implemented 2 column layout, which worked well. Navigation was also excellent - with full context sensitivity. The CSS was clean and concise. I was rewarding an attempt to use more than one sheet - a generic and a set of styles (perhaps that could have been used on different sites).

You didn't attempt EJS. Templating is important in the remainder of the course (and in successor modules) - so it would be worth, perhaps, having a go a the template labs if you have time.

I didn't see any media, but I was able to verify a deployment

Overall I was pleased with the projects - in my categorisation there were 7 outstanding projects, 10 excellent, 6 very good, 4 good and 1 fair. The were graded more or less adhering to the published grading spectrum.

Notify students ☐

Save changes

Save and show next

Reset



Tutors



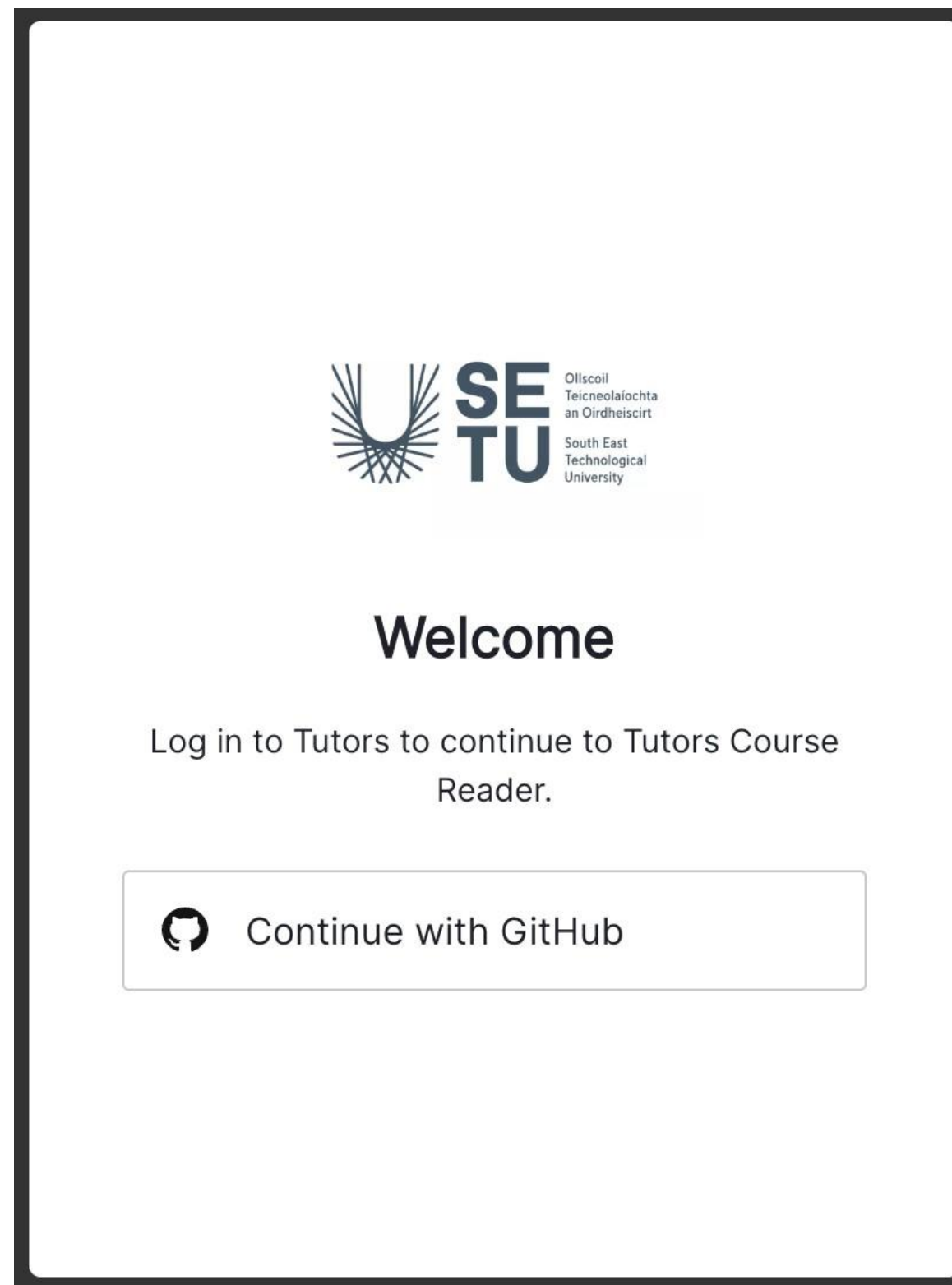
TutorsTime



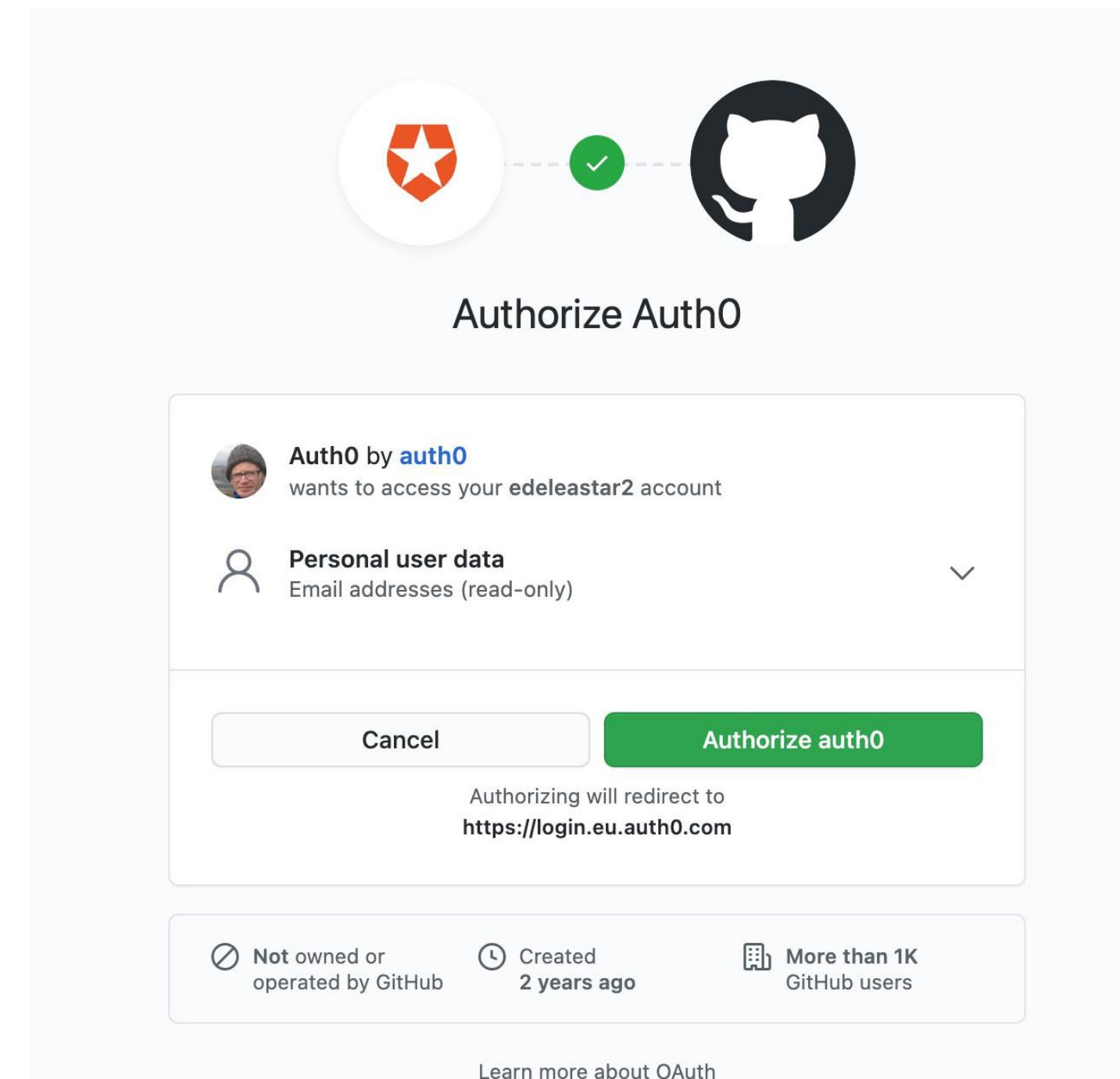
TutorsLive



When accessing module, you will be asked to sign in with Github



... first time you will have to authorise ...



TutorsTime

You have been authenticated via your GitHub credentials and are about to enter a Tutors course. Tutors will record how much time each view is active and send this to the TutorsTime data store. You can view this information via the TutorsTime feature on the course Navigator.

This data is available to you + the instructor of your course, but not to other students. No other data is gathered, nor is this data transmitted anywhere other than the TutorsTime data store.

TutorsLive also uses this data to display of panel of students currently online. You can opt out of this feature from the TutorsLive page.

To learn more about these features please consult:

- [Tutors Time FAQ](#)
- [Tutors Live FAQ](#)

Tutors is an open source application - the data collection component [is here](#).

You will be asked to authenticate again in seven days time.

[Proceed to Course](#)

Github Profile



[Pull requests](#) [Issues](#) [Marketplace](#) [Explore](#)

Your profile picture has been updated. It may take a few moments to update across the site. 

 **edeleastar2**
Personal settings

[Profile](#)
[Account](#)
[Appearance](#) New
[Account security](#)
[Billing & plans](#)
[Security log](#)
[Security & analysis](#)

Public profile

Name

Eamonn de Leastar

Your name may appear around GitHub where you contribute or are mentioned. You can remove it at any time.

Public email

Select a verified email to display

You have set your email address to private. To toggle email privacy, go to [email settings](#) and uncheck "Keep my email address private."

Bio

Profile picture



 Edit



TutorsTime - Labs

T

Web Development

Eamonn de Leastar

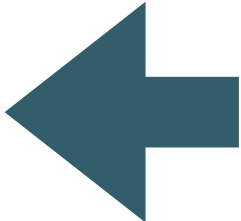
Layout

Labs

Labs Chart

Calendar

User	Github	Total	Date Last Accessed	Lab-00 Editing HTML	Lab-01 HTML Structure	Lab-02a CSS Fonts & Colours	Lab-02b CSS Paths & Links	Lab-03a Layout	Lab-03b Multicolumn	Lab-04 Navigation	Lab-05a Harp & Surge	Lab-05b Templates	Lab-06 Nav Params	Lab-07a Semantic-UI	Lab-07b Play Setup	Lab-08 Playlist 1
Eamonn de Leastar	edeleastar	1337	2/01/2022, 10:12:3	50	8	11	2	16	28	22	22	6	41	2	128	47



Layout

0

Logged in as:

Eamonn de Leastar

Share Presence

View

0

Online

Tutors Live

Tutors Time

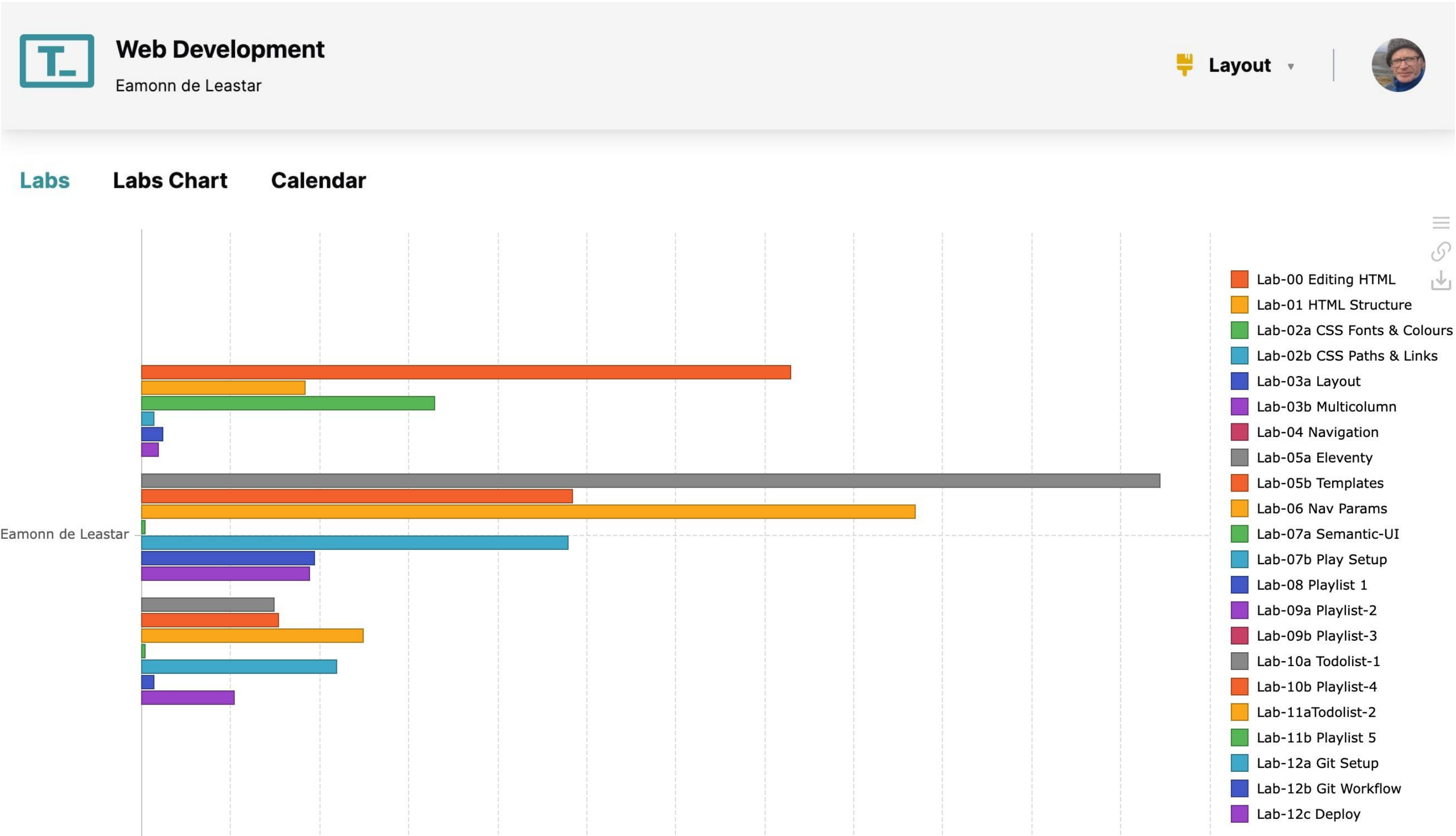
Github Profile

Logout

Number of minutes each lab has been active in your browser (estimate)



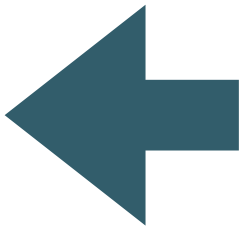
TutorsTime - Chart





TutorsTime - Calendar

	Web Development Eamonn de Leastar														Layout	
Labs	Labs Chart														Calendar	
User	Github	Day	-1: Jan	0: Jan	1: Jan	2: Jan	3: Feb	Reading 1: Feb	4: Feb	5: Feb	6: Mar	Reading 2: Mar	7: Mar	8: Mar	9: Mar	Reading 3: Apr
Eamonn de Leastar																
		Mon	1	4		6	2	109		4	21	2		61	5	18
		Tue	57		12	44	54			46	46					42
		Wed			164	64	6			25	82	51		76	53	
		Thur	2		505	70	327	10	157	192	380	156	258	546	62	13
		Fri	4	1	38	11	19		2	122	302	18	9	111	10	2
		Sat	36		14	7	4				8				138	41
		Sun			16	17			1			6			14	1



Layout

0

Logged in as:
Eamonn de Leastar

Share Presence

View 0 Online

Tutors Live

Tutors Time

Github Profile

Logout

Time online each week of the semester



<https://tutors.dev>

Tutors:An Open Learning Web Toolkit

Open source components & services supporting the creation of learning experiences using web standards. Developed at **SETU**, Waterford, Ireland.

 Create

 Docs

 Source

 Catalogue

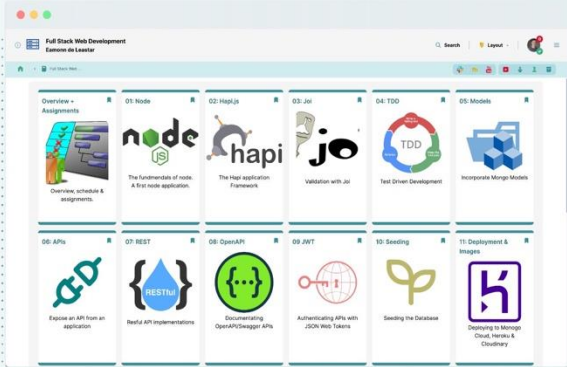
 Live

An Open Learning Web Toolkit

A collection of open source components & services supporting the creation of transformative learning experiences using open web standards.

View Demo

Documentation



The Values of the project

Learner Experience



The **Learner Experience** prioritises web interactions that are **engaging, contextual, linkable, searchable, accessible** and **responsive**. In addition the experience should foster a sense of **community** and **connection** among fellow learners.

Educator Experience



The **Educator Experience** prioritises the creation of a **guided paths** through a curriculum via the creation of learning materials that are **autonomous, structurally aligned, composable, auditable, extensible, versioned** and **independent**.

Developer Experience



The **Developer Experience** prioritises the specification and implementation of **robust, well documented, loosely coupled components & services**, integrated into a **coherent toolkit** open to contributions from **diverse skill sets**.



<https://reader.tutors.dev/course/tutors-project-site>



Tutors Documentation
An Open Learning Web Toolkit



Search



Layout



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 >  Tutors Documen...



Tutors Workshop

First contact with the
Tutors Tool Set

Learner Experience

Consuming a Tutors
Course

Educator Experience

Creating tutors courses

Developer Experience

The Tutors Open Source
Components & Services

TUTORS – NEW Features 2026

- Create a Course (tutors.dev/create)
- Jupyter Notebooks (e.g. run your python scripts within the lab itself)
- Mermaid Diagrams (of all types)
- Marp Talks (instead of pdf)
- New Calendar Format (highlights assignments)
- Layout Menu Options